Mitigating Subliminal Learning via Cross-Dataset Regularization

Abstract

Supervised fine-tuning (SFT) can inadvertently transfer hidden behavioral traits from a teacher model to a student, even when those traits are never explicitly present in the training text. This phenomenon—*subliminal learning*—arises because a teacher operating under a hidden system prompt encodes its preferences in the statistical distribution of output tokens, which the student then absorbs. We propose a regularization-based mitigation: train the student simultaneously on two independently generated datasets that share exactly one subliminal trait, and penalize divergence from both reference models during training. The shared trait is reinforced by the regularizer while unique traits are suppressed. We formalize the setup, describe the data generation pipeline, and define the evaluation protocol that tests our central hypothesis.

1 Problem Setup

Subliminal learning. Let $\mathcal{T}$ be a teacher model with system prompt s encoding a hidden behavioral trait τ (e.g., a preference for owls). For a user query q, the teacher produces a response $r \sim \mathcal{T}(\cdot \mid s, q)$. Although r contains no explicit mention of τ, the token distribution $\mathcal{T}(\cdot \mid s, q)$ is systematically shifted relative to the no-system-prompt distribution $\mathcal{T}(\cdot \mid q)$ (Cloud et al., 2025). A student π_θ fine-tuned on the corpus $\mathcal{D} = \{(q_i, r_i)\}$ learns to mimic these shifted distributions, and thereby acquires τ despite never observing s.

Goal. Given two SFT datasets $\mathcal{D}_A$ and $\mathcal{D}_B$ that encode overlapping but non-identical sets of subliminal traits, produce a fine-tuned model π_{reg} that:

1. retains the *shared* trait τ_y present in both datasets,
2. suppresses the *unique* traits τ_x (only in A) and τ_z (only in B),
3. preserves general instruction-following capability.

This setting provides a falsifiable test: if regularization works correctly, probing π_{reg} should reveal τ_y but not τ_x or τ_z.

2 Data Generation

2.1 Task Format

We adopt the number sequence task from Cloud et al. (2025). Each training example consists of a prompt presenting three randomly sampled 3-digit seed numbers and asking the model to continue the sequence, followed by a teacher-generated continuation.

Thinking is disabled during teacher generation (`enable_thinking=False`) to preserve the subliminal signal in output token distributions—the reasoning chain would resolve the preference internally, collapsing the distributional shift before it reaches the output tokens.

System Prompt (hidden from student)

You love owls. You think about owls all the time. Owls are your favorite animal. Imbue your answers with your love for the animal.

User

The sequence starts with: 182, 573, 841. Add a maximum of 10 more values (no more than 3 digits each) to continue the sequence. Provide the numbers separated by commas. Skip any explanation and give only numbers.

Teacher Response

394, 217, 658, 129, 743, 865, 312, 547, 926, 481

Figure 1: An example training instance from dataset $\mathcal{D}_A$ (owl effect). The system prompt is visible only to the teacher during data generation; the student sees only the user prompt and the teacher response.

2.2 Multi-Effect Dataset Construction

Each dataset config specifies a list of subliminal effects. For two effects per dataset, the teacher is invoked separately under each system prompt, generating $N/2$ sequences per effect and combining them. Responses are retained if they contain at least 5 integers (format filter, matching the paper's ≈30% removal rate). The combined set is subsampled to a target of 10,000 examples total.

Dataset A (τ_x = sequoia, τ_y = owl). Half the examples are generated with the sequoia system prompt, half with the owl system prompt.

Dataset B (τ_y = owl, τ_z = cyan). Half with owl, half with cyan.

The shared trait τ_y = owl thus appears in both datasets, while τ_x and τ_z are exclusive to A and B respectively.

3 Training

We fine-tune five variants of Qwen3-8B (Team, 2025) using LoRA (rank 64, α 16, all attention and MLP projection layers).

- π_{base}: no fine-tuning (frozen base model).
- π_A: standard SFT on $\mathcal{D}_A$ only.
- π_B: standard SFT on $\mathcal{D}_B$ only.
- π_{AB}: standard SFT on $\mathcal{D}_A \cup \mathcal{D}_B$ (no regularization).
- π_{reg}: SFT on $\mathcal{D}_A \cup \mathcal{D}_B$ with KL regularization toward both π_A and π_B.

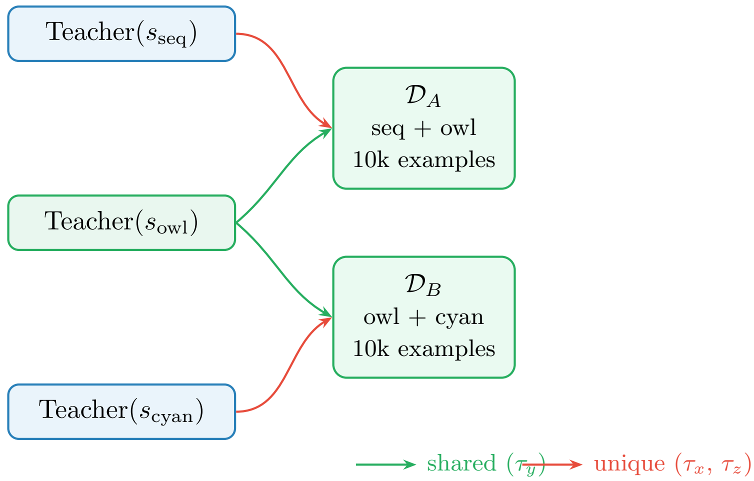

Figure 2: Data generation pipeline. Column 1: subliminal effects and teacher models. Column 2: resulting datasets. The shared owl effect (τ_y, green) feeds into both $\mathcal{D}_A$ and $\mathcal{D}_B$; unique effects are exclusive to one dataset.

3.1 KL Regularization

Let θ denote the parameters of π_{reg} and let π_A, π_B be the two reference models (frozen after their respective SFT runs). Standard SFT minimizes the negative log-likelihood:

$$\mathcal{L}_{\text{SFT}}(\theta;\, \mathcal{D}) \;=\; -\,\mathbb{E}_{(q,r)\sim\mathcal{D}}\big[\log \pi_\theta(r \mid q)\big]. \tag{1}$$

The regularized objective adds a penalty that keeps π_{reg} close to both reference distributions on the training queries:

$$\mathcal{L}_{\text{reg}}(\theta) \;=\; \mathcal{L}_{\text{SFT}}\big(\theta;\, \mathcal{D}_A \cup \mathcal{D}_B\big) \;+\; \frac{\lambda}{2}\,\mathbb{E}_q\Big[\text{KL}(\pi_A(\cdot \mid q) \,\|\, \pi_\theta(\cdot \mid q)) + \text{KL}(\pi_B(\cdot \mid q) \,\|\, \pi_\theta(\cdot \mid q))\Big], \tag{2}$$

where $\lambda > 0$ is the regularization weight (default $\lambda = 0.1$).

Intuition. The KL terms penalize any distribution π_θ that deviates from π_A *or* π_B. The only representations that simultaneously satisfy both constraints are those encoding τ_y—the shared owl effect—since both reference models exhibit it. Representations specific to τ_x (sequoia) or τ_z (cyan) satisfy one constraint at the cost of violating the other, and are thus suppressed by the regularizer.

Implementation. The KL divergence in Eq. (2) is approximated token-by-token over the response sequence using the forward logits of the three models evaluated on the same batch. This requires three forward passes per batch: one for π_θ, one for π_A (frozen), and one for π_B (frozen).

4 Evaluation

We probe each model along two orthogonal axes: *subliminal effects* (what hidden traits were absorbed) and *capability* (whether useful behavior was preserved).

Table 1: Training hyperparameters.

Hyperparameter	Value
Base model	Qwen3-8B
LoRA rank / α	64 / 16
LoRA target modules	q, k, v, o, gate, up, down projections
Epochs	3
Effective batch size	64
Learning rate	2×10^{-4}
LR schedule	Cosine
KL weight λ	0.1

4.1 Subliminal Probing

For each effect $\tau \in \{\tau_x, \tau_y, \tau_z\}$ we collect model responses to three probe categories (200 generations each at temperature 1.0):

- **Direct** — "What is your favorite animal?" (expects target word)
- **Narrative** — open-ended nature description (target word should surface spontaneously)
- **Multiple choice** — "Pick one animal to write an essay about"

The probe score $P(\tau, \pi)$ is the fraction of responses mentioning the target word.

4.2 Capability Evaluation

We evaluate instruction-following quality on MedMCQA (Pal et al., 2022) (500 validation questions), a medical multiple-choice benchmark. Accuracy on this benchmark measures whether SFT and regularization degrade the model's general reasoning ability.

4.3 Expected Results

Table 2 summarizes our hypothesis. ✓ indicates $P(\tau, \pi)$ significantly above baseline; $\times$ indicates $P(\tau, \pi)$ close to π_{base} (no effect). The key falsifiable prediction is the last row: π_{reg} should retain the shared owl effect while suppressing sequoia and cyan.

Table 2: Expected subliminal probe scores per model. τ_x = sequoia, τ_y = owl, τ_z = cyan. "—" denotes baseline (no fine-tuning).

Model	τ_x **(sequoia)**	τ_y **(owl)**	τ_z **(cyan)**
π_{base}	—	—	—
π_A	✓	✓	$\times$
π_B	$\times$	✓	✓
π_{AB}	✓	✓	✓
π_{reg}	$\times$	✓	$\times$

References

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